

Vanguard 3.1

Self-Service Business Intelligence

Making Security Work Measurable

The InfoSec organization at Apple struggled to answer one question "How are we keeping Apple safe?". I led the Business Intelligence (BI) team to build a governed self-service intelligence platform that provided a clear, consistent, and defensible answer using AI for speed and human validation for trust - helping Apple leaders make informed decisions timely, and with confidence.

[full width vg1_homepage.png]

Role and Timeline	Team	Outcome
Lead product designer. Strategy interaction model .governance design. 2025 Q3 - 2026 Q2	1 Designer (me), 4 Developers, 2 Data engineers, 1 PM, 1 SME, partner, security orgs	BI coverage 30% → 75%+ · Escalation calls during incidents cut by more than half

Context

Vanguard Business Intelligence (BI)

The BI team’s charter was to publish performance metrics for senior leadership and partner teams to help make strategic forward looking decisions. They closely worked with each partner team to hand-build metrics - defining the measure, negotiating the formula, working with multiple data engineering teams, publishing, and ongoing maintenance.

Metrics published in Vanguard was a single source of trusted truth for Apple leadership.

[vg2_mttdd_metric.png] [vg2_programs.png]

Problem

The small BI team could only cover ~30% of the security teams. Others ended up creating their own dashboards

Executives had no defensible way to answer "how are we keeping Apple safe?"

to share with leadership directly. Executives were struggling for one holistic view they could trust. My design goals:

1. Increase BI coverage - through simplified or faster onboarding.
2. Design and implement a holistic view for leadership - using a top-down results map.
3. Improve the platform interface to better the dashboards of non-BI teams.

[vg3_results_map.png] [vg3_impact_metrics.png]
[vg3_incident_management.png]

Research

Metrics went stale, and anomalies unexplained.

I started with a design sprint with the BI team to align on goals and map how a metric actually gets made. Then I ran interviews on both ends of the pipe. I found onboarding was slow (avg. ~2 months) primarily due to iteration between the two teams. Also, different teams had a variety of dashboards - using varied metrics across the board. But my conversations with partner teams and executives uncovered two underlying challenges:

Published metrics went stale soon. Teams had to spend a lot of time continuously working with the BI team to maintain their metrics, and back-and-forth caused more friction than help.

Anomalies were displayed, but not explained. So, when a metric spiked, there was a flurry of questions, check ins, and chaos as leadership wanted to understand the full picture.

[vg5.png] [vg6.png]

Strategy

The team had two plans. I argued for one.

The design sprint I organized with the core team went like this:

- **Director:** More BI headcount could not solve the bottleneck problem, since teams are looking for autonomy and control.
- **Me, Director, Dev Lead:** Lets expand to a self-service platform!
- **PM:** That will dilute the trust factor we gained over years. The design challenge was

If anyone can publish, why would an executive believe anything?

We also collectively agreed to bring an AI powered anomaly, detection, and analysis - and the reframed design challenge was:

"how might we make governance and trust signals visible in a scalable business intelligence platform?"

[vg61.png][vg62.png]

Design 1/4

Quality gates built into the onboarding flow

Rather than reviewing quality after submission, I baked checks in the authoring path. The self-serve flow asks the same questions the BI team used to ask in review meetings:

- Is there an explicit, unambiguous formula for this measure?
- Are all contributing data sources declared?
- What does "good" look like, and against what baseline?

[vg_contribute_1.png through vg_contribute_6.png, like a fade in auto carousel]

Design 2/4

Always-attached trust tiers and data lineage

We started with Gold/Silver/Bronze states, but initial tests revealed attitudinal biases - lower tiers felt like something to hide, as opposed to a step to climb. We changed it to actual state which describes the provenance instead of status.

Tier	Meaning
BI Led	Defined and built by the central BI team
BI Approved	Built by the owning team, reviewed and signed off by BI
Self-Service	Built and published by the owning team

Each metric also had their data lineage always visible, so a viewer can determine how much they should trust a measure.

[vg8_trust_tiers.png]

[vg8_mttdd_anomaly_data_lineage.png]

Design 3/4

Human-in-the-loop anomaly analysis

When a measure broke its control limits, an automated analysis was drafted by an agent and a notification sent to the directly responsible individual (DRI). Once edited and approved by the DRI, the analysis was available on the dashboard.

Trade-off: initially, the team wanted to hide automated analysis unless they were verified by a human. But the executives wanted to avoid another bottleneck of approval, and insisted on viewing AI analysis as well. So we added a confidence score along with reasoning. Most importantly, I made it very clear visibly when somebody was looking at automated analysis.

[All 3 vg9....png, side by side]

Design 4/4

The executive

The strategy was to have a single yard stick for every measure. We used a statistical process control chart, so we are able to distinguish normal variation from a genuine signal. This made it easy for the executives to

surface

react to false positives like “we are off our average this week”.

[vg1_homepage.png]

[vg7_homepage_critical_services.png]

Outcome

The IKEA effect

During pilot, I noticed team leads were excited with the new-found autonomy. They were taking care of it like they own it. We crossed 30% → 75% coverage in 3 months.

Teams prioritized submitting well crafted measures for their teams, which in turn made it easy for the BI team to review and approve within days. Average on boarding time reduced from ~2 months → <5 days

Escalation calls during anomalies reduced by more than half (as teams reported, survey June 2026).

Trust doesn't scale through review capacity. It scales when it becomes a visible property of the artifact. - Apple InfoSec BI team PM